

Worked Solution (Formula Solution)

a. What is the present value of your windfall?

The present value of a cash flow stream is:

$$PV = C_0 + \frac{C_1}{(1+r)} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_N}{(1+r)^N}$$

where PV is the present value, C_n is the cash flow at date n , r is the interest rate, and N is the date of the last cash flow in a stream of cash flows.

Therefore,

$$PV = \frac{\$30,000}{(1+0.047)} + \frac{\$60,000}{(1+0.047)^2} + \frac{\$90,000}{(1+0.047)^3} = \$161,802.98$$

The present value of your windfall is \$161,802.98.

b. What is the future value of your windfall in 3 years (on the date of the last payment)?

To find the future value of a cash flow, use the following formula:

$$FV_n = C \times (1+r)^n$$

where FV_n is the future value on date n .

The future value of your windfall is found by finding the future value of the answer in part (a):

$$FV_3 = \$161,802.98 \times (1+0.047)^3 = \$185,706.27$$

The future value of your windfall in 3 years is \$185,706.27.

